INVESTIGATOR BROCHURE
PrP-siRNA

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**Elsewhere called 2439-s4 or 2439-exNA.*

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Confidentiality Statement

This is a confidential document for the sole information and use of the investigator's team and the IRB/IEC.

Abbreviations

AE	Adverse Event
ASO	Antisense oligonucleotide
CJD	Creutzfeldt-Jakob disease
CSF	Cerebrospinal fluid
di-siRNA	Divalent short interfering RNA
dpi	Days post-infection
ED_{50}	Median effective dose
ELISA	Enzyme-linked immunosorbent assay
exNA	Extended Nucleic Acid
FDA	Food and Drug Administration
FFI	Fatal familial insomnia
GLP	Good Laboratory Practices
GSS	Gerstmann-Straussler-Scheinker disease
ICH	International Conference on Harmonization
LP	Lumbar puncture
MRC-PDRS	Medical Research Council Prion Disease Rating Scale
PD	Pharmacodynamic
PK	Pharmacokinetic
PrP	prion protein
RT-QuIC	Real-time quaking-induced conversion
siRNA	short interfering RNA

Executive Summary

PrP-siRNA is a small interfering RNA (siRNA) drug candidate designed to lower prion protein (PrP) in the human brain. It is indicated for the treatment of adults either with symptomatic prion disease or at high genetic risk for prion disease.

PrP is the cause of all cases of prion disease, regardless of etiology (sporadic, genetic, acquired) or subtype (CJD, FFI, GSS, or other). PrP is present in the brains of all mammals, but in disease, it misfolds, and the misfolded conformer is self-templating, leading to exponential increase in the amount of misfolded PrP, culminating in rapid neurodegeneration and death[1]. All available lines of evidence including neuropathology, biochemistry, human genetics, and animal genetics agree that PrP is the root cause of disease[1,2]. Prion disease naturally afflicts many other mammals besides humans, and is well-modeled in wild-type mice, where infection with prions results in fatal neurological disease within 6 months[3]. Animals lacking PrP are completely resistant to prion infection, and reduction of PrP by heterozygous genetic knockout or by pharmacologic lowering dramatically delays disease onset and slows disease progression[2,4,5].

PrP-siRNA[6] is an oligonucleotide drug consisting of two covalently linked sense strands, each hybridized to an antisense strand that is complementary to the human *PRNP* mRNA, which encodes PrP. The antisense strands are loaded into the RNA-induced silencing complex (RISC) and cause degradation of the *PRNP* mRNA, lowering steady state levels of PrP. PrP-siRNA does not cross the blood-brain barrier and must be dosed directly into CSF. In humans, it will be administered by intrathecal injection.

PrP-siRNA was discovered by screening siRNA molecules in human cultured cells and in transgenic mice expressing human PrP, and was selected as the most potent PrP-lowering siRNA in these screens[6]. In transgenic human PrP mice, a single dose of PrP-siRNA at 35 – 52 µg can lower whole brain hemisphere PrP by 50%; these dose levels, based on CSF volume scaling, are expected to correspond to 59 – 169 mg in humans.

Homozygous PrP knockout is known to be tolerated in mice[7], cattle[8], and goats[9], and naturally occurring heterozygous PrP knockout alleles have been found in healthy humans[10,11]; thus, on-target toxicity from PrP lowering is not expected. Nonclinical GLP toxicology studies were conducted in rats and dogs, where PrP-siRNA is not sequence-matched to PrP; there is no on-target pharmacology, but off-target toxicity can still be assessed. Toxicology findings were non-adverse up to the maximum dose tested, which was 3 mg in rats and 200 mg in dogs. In vitro studies yielded no evidence of drug-drug interactions or genotoxicity.

PrP-siRNA is supplied as 5 mL vials formulated at 10 mg/mL. The target dose volume is 20 mL across all dose levels; during dose escalation, the product will be diluted to target dose in Elliotts B prior to injection.

Two clinical trials are planned, one in symptomatic individuals diagnosed with prion disease and the other in pre-symptomatic individuals at risk for genetic prion disease. In each case, the trial will be 50 mg, 100 mg, and 200 mg single ascending dose, with follow-up LPs after dosing to determine target engagement by measuring CSF PrP. The goal of these trials is to determine whether the drug is reasonably safe based on the primary endpoint of adverse events, and to determine the pharmacokinetics and pharmacodynamic impact of the drug in each patient population.

Introduction

PrP-siRNA is a PrP directed small interfering RNA indicated for the treatment of adults diagnosed with, or at genetic risk for, prion disease.

Rationale for PrP lowering. PrP is the established cause of all cases and all known forms of prion disease, including Creutzfeldt-Jakob disease (CJD), fatal familial insomnia (FFI) and Gerstmann-Straussler-Scheinker disease (GSS). Prion disease naturally afflicts many mammals other than humans, and is exceptionally well-modeled in mice, where infection with exogenous prions results in fatal disease within 6 months. PrP lowering is a clear therapeutic hypothesis for treating prion disease: in mice, PrP lowering either by heterozygous genetic knockout or by treatment with a PrP-lowering antisense oligonucleotide (ASO) delays the onset and slows the progression of prion disease, increasing overall survival. Treatment is dose-responsive, where deeper lowering of PrP results in longer survival time. A PrP-lowering ASO, ION717, is now being tested in a Phase I trial sponsored by Ionis Pharmaceuticals, with expected primary completion in July 2025; no results are known at this time. Prion disease is a serious unmet medical need; there is currently no standard of care, and no other experimental trials for prion disease globally.

Rationale for safety of PrP lowering. PrP knockout is known to be tolerated in mice[7], goats[9], and cattle[8], and humans heterozygous for PrP loss-of-function mutations are healthy[10,11]. Post-natal depletion of PrP and lowering of PrP with ASOs have been found safe in mice[5,12,13]. The only known phenotype of PrP knockout is a phenotypically very mild peripheral neuropathy, found only in aged (60 week old) homozygous knockout animals, not in heterozygotes or young animals, and only in peripheral nerves, not in the CNS[14]. Because this phenotype appears to

require deeper than 50% PrP lowering for more than 1 year, it is not likely to be observed in this clinical trial of a single dose PrP-siRNA.

Rationale for divalent siRNA. PrP-siRNA belongs to a new chemical class of oligonucleotide drugs called divalent siRNA[15]. Like ASOs, divalent siRNAs are sequence-targeted to an RNA molecule of interest, and cause the cleavage of that target RNA. Whereas ASOs exhibit approximately 12 weeks of pharmacologic activity in vivo and can achieve ~50% PrP lowering, divalent siRNAs have been shown active out to 6 months and can achieve deeper lowering, with as little as 17% residual PrP measured in mouse whole brain hemisphere[6].

Pharmacodynamic biomarker. This divalent siRNA is designed to lower the human *PRNP* gene RNA, resulting in lowering of PrP expression in the brain. In rats and in cynomolgus monkeys, lowering of brain PrP is reflected in CSF[16,17]. Therefore, it is expected that the pharmacodynamic impact of this drug can be read out by measuring CSF PrP. PrP is a sticky protein that is readily adsorbed onto polypropylene surfaces[18]. In order to achieve reliable quantification of PrP, it is essential to follow our published protocol[18], using identical containers and an identical amount of pipetting and perturbation to all samples, and adding detergent (0.03% CHAPS) to aliquots intended for PrP analysis. Participants will undergo a screening and baseline CSF collection, and subsequent CSF collection at 4 week intervals after drug administration in order to monitor PrP lowering. CSF samples collected in this study will be compared to individual baselines for each individual as well as to samples collected in separate observational studies of pre-symptomatic at-risk individuals and of symptomatic subjects diagnosed with prion disease.

Patient population. PrP-siRNA will be tested in two clinical trials, one in symptomatic patients and one in pre-symptomatic at-risk individuals.

For symptomatic individuals diagnosed with prion disease, inclusion criteria require that patients meet the CDC diagnostic criteria for prion disease, and have either a positive CSF RT-QuIC test or a positive *PRNP* gene mutation test. These two molecular biomarkers in combination are highly sensitive for both sporadic and genetic prion disease, while virtually guaranteeing the exclusion of any patient without prion disease[19]. Divalent siRNA lowers the PrP RNA, leaving the protein itself to degrade on its own half-life of ~5 days[20]. This means there is a lag time before full pharmacodynamic effect can be achieved, and indeed, mouse studies of ASOs were unable to show any benefit in animals treated fewer than 3 weeks before death[5]. In this rapidly progressive disease it is essential to treat patients with sufficient lifespan remaining to potentially benefit from the drug, and because the disease is rapidly debilitating, it is essential to focus on patients who still have quality of life left to preserve. For both of these reasons, inclusion is limited to patients with an MRC-PDRS[21] of ≥15.

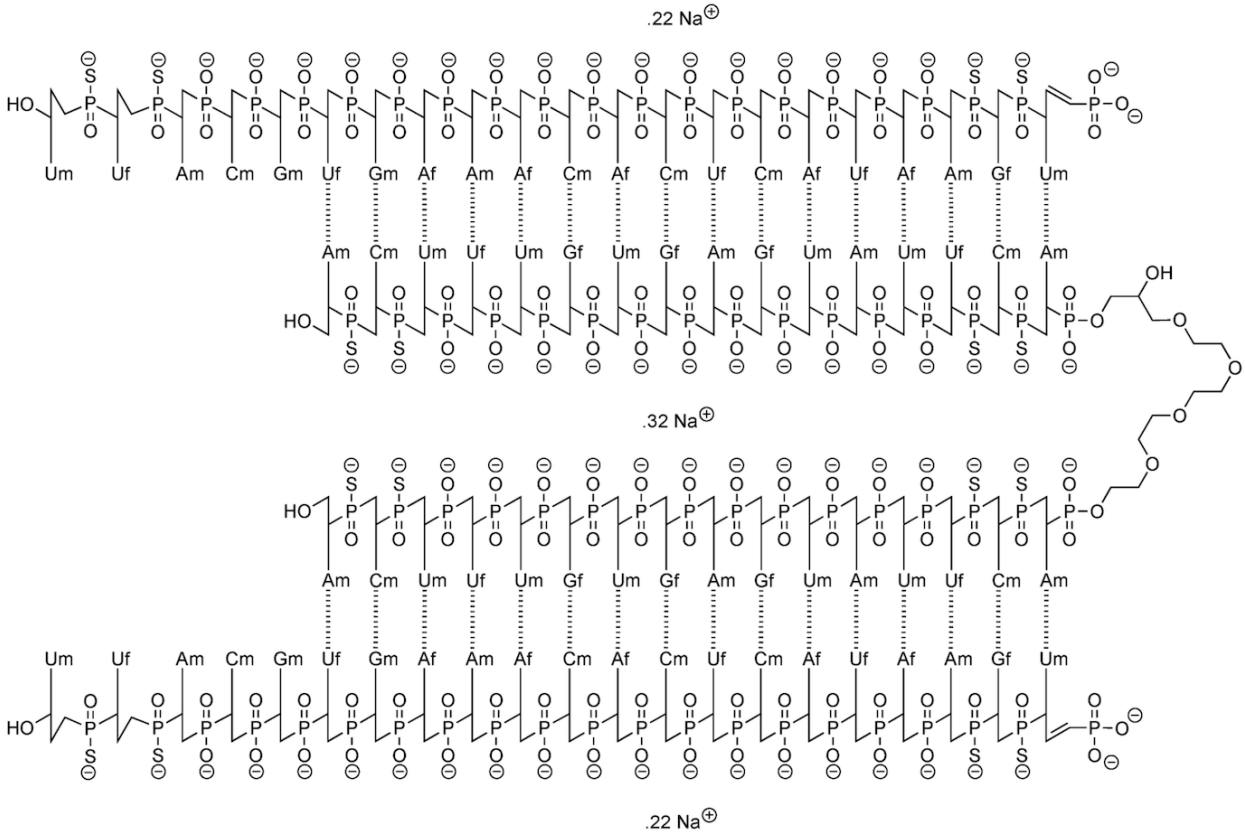
For pre-symptomatic individuals at risk for prion disease, inclusion criteria require a positive *PRNP* gene mutation test for a highly penetrant[22] mutation, without clinical symptoms of prion disease.

Physical, Chemical, and Pharmaceutical Properties and Formulation

Drug Substance

The full structure of PrP-siRNA is shown in Figure 1 below. Note that PrP-siRNA is the same molecule referred to elsewhere as 2439-s4 or 2439-exNA. It consists of two covalently linked sense strands, each hybridized to an antisense strand. The sequence of the antisense strand is UGAAUACUCACAAAGUGCAUU. The antisense strand is loaded into the RNA-induced silencing complex (RISC) where it catalyzes the cleavage of the *PRNP* mRNA, which is sequence-matched in its 3' untranslated region to the core GAAUACUCACAAAGUGCA sequence of the antisense strand. The backbones of the siRNA strands are stabilized with several chemical modifications: phosphorothioate (PS), 2'-O-methyl (2'OMe or m in the below diagram), 2'-Fluoro (2'F or f in the below diagram), extended nucleic acid (exNA), and 5'-vinyl phosphonate (5VP), which variously assist with cellular uptake, RISC loading, and/or exonuclease resistance.

Figure 1. Structure of PrP-siRNA



The Drug Substance (DS) was synthesized by Hongene Biotech (Shanghai, China) as a lyophilized white powder, and subsequently formulated into Drug Product (DP) by Argonaut Manufacturing Services (Carlsbad, CA).

Drug Product

The DP is a clear, sterile injectable liquid formulated at 10 mg/mL in 5.3 mL vials with guaranteed 5 mL extractable volume PrP-siRNA is formulated in saline with magnesium ions in a 25:1 molar ratio to the divalent drug molecule.

Table 1. Clinical formulation of PrP-siRNA Drug Product

ingredient	concentration
PrP-siRNA	10 mg/mL
potassium dihydrogen phosphate	0.144 mg/mL
sodium chloride	9.058 mg/mL
sodium phosphate dibasic heptahydrate	0.796 mg/mL
magnesium chloride hexahydrate	2.114 mg/mL

The incorporation of divalent cations such as magnesium or calcium is necessary to prevent ionic imbalance[23–25]. Absent these cations, oligonucleotides can cause seizures believed to be due to high-affinity divalent cation binding sites on the nucleic acid backbone depleting calcium from CSF. The ratio of 25:1 was established as safe based on studies in both waking and anesthetized animals[23,25] and was well-tolerated in our GLP toxicology studies. The safety of this dose of intrathecal magnesium is supported by our toxicology studies, which also utilized the 25:1 magnesium:drug ratio. In dog toxicology studies, the maximum dose of 200 mg contained the same amount of Mg^{2+} as the human 200 mg dose will contain, while being administered to a species with only 12% the CSF volume of humans[26].

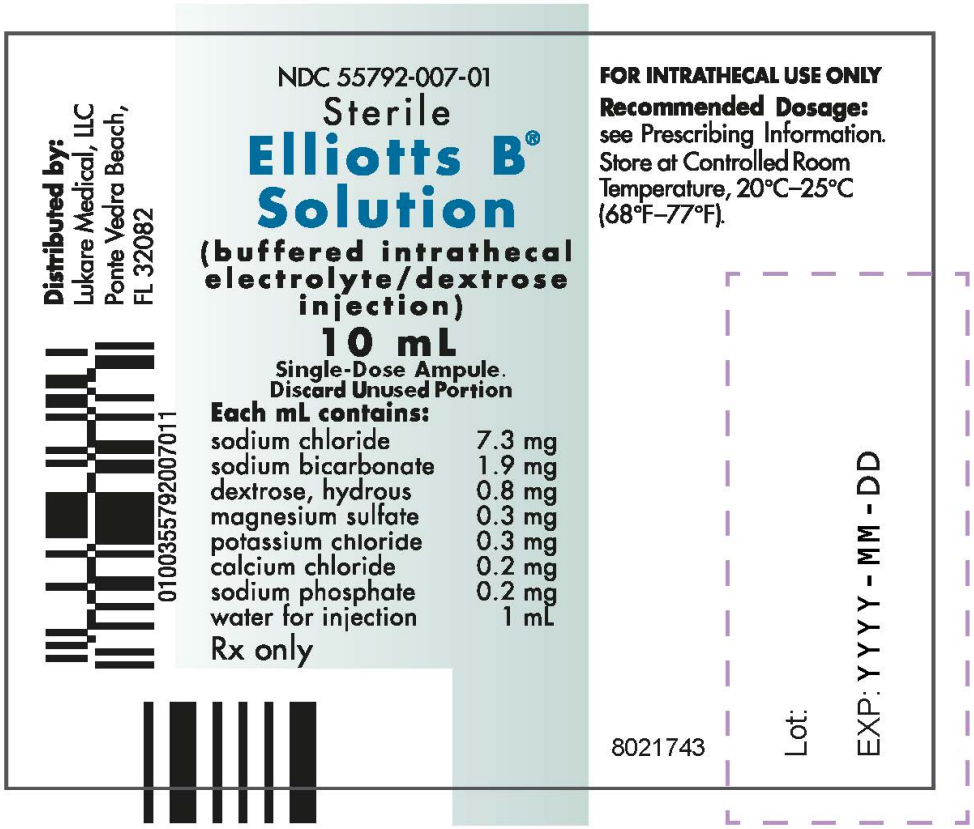
The planned doses for this clinical study are 50, 100, and 200 mg. Vials will be shipped as needed from a GMP storage facility to the MGH Research Pharmacy. The starting dose is 50 mg, with planned escalation to 100 and 200 mg (see dose escalation scheme below). Regardless of the dose level in milligrams, the dose volume is 20 mL, because both nonclinical and clinical studies have suggested that higher dosing volumes result in broader brain biodistribution of intrathecal oligonucleotides. Thus, for the 50 mg and 100 mg doses, vials of drug product will be diluted in Elliott’s B (Lukare Medical) at the hospital's Research Pharmacy. Regardless of dose level, a 20 mL syringe containing the indicated amount of drug will be provided by the Research Pharmacy to the investigator. The study drug solution will be administered to the patient within 4 hours of preparation after storage in the syringe at room temperature.

For the 50 mg and 100 mg doses, we intend to administer these doses in 20 mL dosing volume, even though the Drug Product is formulated at 10 mg/mL. In order to achieve this, we plan to dilute the Drug Product into Elliott's B, an approved intrathecal solution. The final injection will thus include 15 mL of Elliotts B at the 50 mg dose and 10 mL at the 100 mg dose. The inactive ingredients in Elliotts B are shown in Table 2.

Table 2. Inactive ingredients in Elliotts B, proposed diluent for 50 mg and 100 mg doses of PrP-siRNA.

ingredient	concentration
sodium chloride	7.3 mg/mL
sodium bicarbonate	1.9 mg/mL
anhydrous dextrose	0.8 mg/mL
magnesium sulfate	0.3 mg/mL
potassium chloride	0.3 mg/mL
calcium chloride	0.2 mg/mL
sodium phosphate	0.2 mg/mL

The labeling for Elliotts B is shown below.



Nonclinical Studies

PrP-siRNA is designed to lower the human *PRNP* mRNA, encoding PrP, resulting in a reduction of PrP expression. As for all siRNA drugs, its activity is mediated by the antisense strand of the siRNA being loaded into the RNA induced silencing complex (RISC) and binding the target mRNA by Watson-Crick base pairing. PrP-siRNA is intended to be dosed into CSF, and all nonclinical studies were conducted by dosing into CSF. For mice, intracerebroventricular (ICV) injections were used, while in rat and dog, intrathecal (IT) injections were used, consistent with the clinical route of administration. PrP-siRNA was studied first in transgenic mice expressing human PrP to characterize its pharmacodynamic effects, and then subjected to GLP toxicology studies in rats and dogs. Drug-drug interaction and genotoxicity studies were also conducted. All of these studies are summarized below.

Nonclinical pharmacology

PrP-siRNA was discovered through screening of siRNAs in human cultured cells and in transgenic mice expressing human *PRNP* to determine the siRNA sequences and chemical backbones providing the deepest lowering of the *PRNP* RNA transcript and PrP protein[6]. In human PrP transgenic mice dosed intracerebroventricular (ICV) injection, both the sequence and backbone of PrP-siRNA were determined to be superior to other compounds tested. At high doses in mice (174 or 348 µg), suppression to as little as 28% or 17% residual PrP in whole brain hemisphere was observed, and activity was observed out to 4-6 months. In dose response, the median effective dose, or dose to achieve 50% lowering of *PRNP* mRNA), was determined to range from 5 to 18 µg across 6 brain regions evaluated. Based on CSF volume scaling (a factor of 3,250 from mouse to human), 18 µg may correspond to 59 mg in a human. Because PrP-siRNA is not sequence matched for mouse *Prnp*, a surrogate compound with the same chemical backbone but different nucleotide sequence targeting mouse *Prnp* was utilized to study the effects of PrP lowering by this modality in a disease model, wild-type mice infected with the RML strain of prions. Survival time was increased 2.7-fold with early (pre-symptomatic) chronic divalent siRNA treatment, and survival was increased by 64% even with a single dose administered at a symptomatic disease stage.

Efficacy studies were also conducted in a disease model, using a tool compound with a chemical backbone like that of PrP-siRNA but with a sequence specific to the mouse *Prnp* gene. These studies utilized a standard disease model: wild-type mice inoculated intracerebrally with RML prions. Treatment every 4 months, beginning at a disease stage with neuropathology but before frank symptoms, with a dose that lowers PrP to 49% residual, yielded a 2.7-fold increase in

survival time (from 165 to 442 days on the median). Treatment with a single dose upon the onset of symptomatic disease indicated by weight loss yielded a 64% increase in survival time (from 164 to 270 days on the median). These results confirmed that PrP lowering by this modality — just as for genetic knockout or ASO-mediated PrP lowering — is effective against disease.

Pharmacokinetics and Product Metabolism in Animals

PrP-siRNA was detected in the CSF for most dogs and all rats on Day 2 (~24 hours) following siRNA PrP-siRNA administration, although the concentrations were highly variable. However, even in animals where siRNA PrP-siRNA was not detected in CSF, exposures in spinal cord and brain confirmed that PrP-siRNA was delivered to the intrathecal space.

Evaluation of PrP-siRNA tissues levels showed the highest concentration on Day 2 in the liver. siRNA PrP-siRNA was observed in the lumbar and cervical spinal cord, with higher concentrations observed in the lumbar spinal cord which correlates to the site of administration. In the brain, siRNA PrP-siRNA was observed at the highest concentration in the frontal cortex, hippocampus, and cerebellar cortex, but widespread exposure was noted with lower concentrations noted in the thalamus, putamen, and caudate. Generally, PrP-siRNA remained present in liver, kidney, spinal cord, and brain at Day 29, but at lower concentrations compared to Day 2.

Toxicology

Complete (homozygous) PrP knockout, by genetic deletion of the PrP gene, is known to be tolerated in mice, goats, and cattle. Humans heterozygous for PrP loss-of-function mutations are healthy. Given the existing evidence for safety of PrP lowering, in consultation with FDA, GLP toxicology studies were conducted in both rats and dogs, neither of which are pharmacodynamically relevant species but which permit assessment of off-target toxicity.

The same GMP batch of PrP-siRNA that will be used clinically in Phase I trials was used in the GLP IND-enabling studies (see section 3.2.S). The completed toxicology studies include single dose studies in rats and dogs, with necropsies at 24 hours and 28-days. The intrathecal (IT) route of administration was used for the nonclinical safety studies because it is the proposed clinical route. No adverse toxicity was identified in the completed siRNA PrP-siRNA GLP toxicology

studies in rats and dogs. The NOAEL in rats was 3.0 mg and the NOAEL in dogs was 200 mg, the highest dose tested in each species. In 4/8 dogs receiving 200 mg, and 1/8 dogs receiving 60 mg, gliosis was observed at 24 hours post-dose, but this was not associated with any necrosis or degeneration except focally at the injection site in 1 animal at the 200 mg dose. No gliosis was observed at the 28 day post-dose recovery timepoint. The gliosis was classified as non-adverse by the pathologists. No gliosis or other pathological findings were observed in rat. Safety pharmacology endpoints (i.e., respiratory, cardiovascular and neurobehavioral endpoints) were incorporated into the dog GLP study. No test article related effects were observed for the safety pharmacology endpoints measured.

PrP-siRNA was evaluated for potential to induce or inhibit a series of CYPs using cultures of cryopreserved primary human hepatocytes, as an inhibitor of the human transporters P-gp, BCRP, OATP1B1, OATP1B3, OAT1, OAT3, OCT2, MATE1 and MATE2-K, and a substrate of human transporters P-gp and BCRP. No significant effects were found in any of these studies.

PrP-siRNA was evaluated for genotoxic potential in alignment with Option 1 of ICH S2 (R1) (2011). The studies conducted were the in vitro bacterial reverse mutation assay (GLP) and the vitro micronucleus test with the Chinese hamster ovary derived CHO K1 cells (GLP). The highest concentrations of PrP-siRNA used for the in vitro assays, 5000 mcg/plate, were the maximum dose level recommended per ICH S2A(R1). Genotoxicity studies did not identify notable liabilities of PrP-siRNA; the compound was negative in both the bacterial reverse mutation assay (i.e., non-mutagenic) and in the chromosomal aberration assay in cultured Chinese hamster lung fibroblast with or without metabolic activation (i.e., non-clastogenic).

Effects in Humans

PrP-siRNA has not been administered to humans.

Summary of Data and Guidance for the Investigator

The following adverse events have been reported following administration of other oligonucleotide therapeutics and may represent potential risks:

- Immune-mediated effects
- Hepatotoxicity
- Renal toxicity
- Thrombocytopenia
- Hydrocephalus

Monitor for symptoms and signs that may be clinical manifestations of these types of events.

GLP toxicology studies indicate that PrP-siRNA is well tolerated at intrathecal doses up to 200 mg in dog and 3 mg in rat. Scaled based on species-specific CSF volume, these correspond to doses ~8 times higher than the maximum proposed human dose of 200 mg.

The most likely drug- or procedure-related complications are those related to lumbar puncture or drug related headaches or injection site reactions.

The only known on-target side effect of lowering PrP would be a mild peripheral neuropathy, but because this has only been observed in homozygous knockout animals more than 1 year old, it is not likely to be observed in this trial. Improperly formulated oligonucleotides injected into CSF have been shown to cause seizures, but our drug product was formulated with divalent cations to prevent this phenotype, so seizures are not expected.

The most likely adverse events in this trial are related to disease progression, up to and including death, as this is a uniformly fatal disease. There are not yet any human data supporting efficacy of PrP-siRNA, and based on animal studies, a single dose of PrP-siRNA is not expected to fully halt the disease process. Therefore, it is expected that all symptomatic patients enrolled in the symptomatic trial will experience disease progression and die of prion disease.

No drug interaction studies in humans have been conducted with PrP-siRNA to date.

No data regarding overdose are available.

PrP-siRNA should not be administered to patients who are pregnant or to mothers who are breast-feeding.

Sufficient documentation regarding use of PrP-siRNA in pregnant women is lacking. Reproductive toxicity studies of PrP-siRNA in animals have not been performed. Women of childbearing potential (defined as < 2 years after last menstruation and not surgically sterile) not willing to use highly effective form of contraception such as hormonal birth control or intrauterine device during study participation and for at least 6 months thereafter or have a partner who is sterile (i.e., after vasectomy) should not be exposed for PrP-siRNA In the upcoming trial male patients should refrain from sperm donation and ensure use of highly effective form of contraception with their partners if they are of childbearing potential during trial treatment and up to 6 months following the last dose of PrP-siRNA.

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